

Sai Venkat Gunda

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SUMMARY

Advanced Controls and ML Engineer with 4 years of industry experience. Specialized in **Reinforcement Learning, Modern Control and Machine Learning**. High agency. Always curious. Currently targeting roles in the **Physical AI** space that require a deep mix of physical modeling, simulation, and applied ML.

SKILLS

Languages: Python, C++ **Modelling and Simulation:** MATLAB, Simulink, Gazebo, MuJoCo
ML & Robotics: ROS2(Kilted Kaiju), LeRobot, stable-baselines3, PyTorch, Scikit-learn, Reinforcement Learning(PPO, SAC, DQN), Imitation Learning(ACT, Diffusion Models)

PROFESSIONAL EXPERIENCE

- **Jaguar Land Rover** Bengaluru, India
Advanced Controls & ML Engineer May 2024 - Present
 - **Virtual Sensors:** Engineered and deployed industrialized ML models to virtualize thermal sensors achieving **96%** real time **prediction accuracy** in production vehicles.
 - **ML Data Pipelines:** Built an end-to-end ML pipeline to create Virtual Sensors starting from data collection, pre-processing, filtering, modelling and validation. Reduced the complete delivery cycle from **1 week to 4 hours**.
 - **Engineering Lifecycle:** Managed the complete delivery of Hypervisor feature enabling a **range benefit of 10+ miles** from planning, prototyping, industrialization, validation and delivery.
 - **State Estimation(EKF):** Developing an Extended Kalman Filter for the refrigerant system to replace existing Virtual Sensors with grey-box state estimation techniques for better explainability and reliability of predictions.
 - **Model Predictive Control(MPC):** Currently developing a deployable Model Predictive Controller architecture for vehicle refrigerant systems to optimize between comfort, performance, component durability and efficiency.
- **Jaguar Land Rover** Bengaluru, India
Structural CAE Analyst July 2022 - May 2024
 - **Reduced Order Modelling:** Developed reduced order 1D frontal crash modelling techniques providing quick verification of design ideas in seconds with an **accuracy of 80%**

PROJECTS

- **Intrinsic AI for Industry Challenge** Bengaluru, India
Robotics challenge by Intrinsic, Google Apr 2026 - Present
 - **Robot Learning Pipeline:** Built an end-to-end **robot learning pipeline** in Gazebo, generating 1,000+ training trajectories with kinematic perturbations and domain randomization to ensure robust policy training.
 - **ACT Policy:** Fine-tuned an Action Chunking with Transformers (ACT) policy using a **ResNet18** backbone via **LeRobot**; achieved **sub-10mm positioning accuracy** but identified failure modes during contact-rich insertion.
 - **Diffusion Policy:** Implementing a diffusion-based control policy to specifically resolve ACT's contact dynamic failures, targeting **high-precision, sub-millimeter port insertion**.
- **Autonomous All-terrain Mars Rover** IIT Madras
Team and Traversal Lead Jan 2018 - May 2021
 - **5 DOF Manipulator:** Developed a **5 DOF serial manipulator** capable of lifting **5 kg** with a compliant gripper. Developed closed loop control systems for the manipulator using Inverse Kinematics for robust manipulation
 - **All-Terrain Traversal System:** Developed a rocker bogie suspension system for all terrain travel of autonomous rovers capable of climbing 45 cm heights and 30° slopes

EDUCATION

- **Indian Institute of Technology Madras** Chennai, India
Dual Degree(B.Tech + M.Tech) Honours in Mechanical Engineering; GPA: 9.14 Aug. 2017 - May. 2022